

Question	Answer	Marks
3(a)	(resultant) force proportional/equal to rate of change of momentum	B1
3(b)(i)	$\rho = m / V$	C1
	$V = \pi \times (7.5 \times 10^{-3})^2 \times 13 \times 0.2 (= 4.59 \times 10^{-4} \text{ m}^3)$	A1
	$m = \pi \times (7.5 \times 10^{-3})^2 \times 13 \times 0.2 \times 1000 = 0.46 \text{ kg}$	
3(b)(ii)	1. $(\Delta)p = (\Delta m)v$	C1
	$(\Delta)p = 0.46 \times 13$ $= 6.0 \text{ N s}$	A1
	2. $F = 6.0 / 0.20$ $= 30 \text{ N}$	A1
3(b)(iii)	force on water (by rocket/nozzle) equal to force on rocket/nozzle (by water)	M1
	in the opposite direction	A1
3(b)(iv)	1. mass = $0.40 + 0.70 - 0.46 = 0.64 \text{ kg}$	A1
	2. acceleration = $[30 - (0.64 \times 9.81)] / 0.64$ or $30 / 0.64 - 9.81$	C1
	$= 37 \text{ m s}^{-2}$	A1

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4(c)	The first two lines of the poem are written in the first person singular, 'I' and 'me'. This suggests that the speaker is a single individual, possibly a woman, who is addressing the reader directly. The use of 'I' and 'me' creates a sense of intimacy and immediacy, drawing the reader into the speaker's world. The speaker's tone is one of longing and desire, as evidenced by the repeated use of the word 'want' in the first two lines. The speaker expresses a desire for the reader to 'be a part of me', which could be interpreted as a desire for love, companionship, or a shared experience. The overall effect of the first two lines is to establish a personal and emotional connection between the speaker and the reader, setting the stage for the rest of the poem.	B1